

DFS504 The Practice of Big Data and Analysis in the Financial Industry

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Spring 2025 (Jan. 2026-June 2026)

Semester Project: Big Data and AI-Enabled Financial Analytics Solution

1. Project Team Missions

Students will form a team of 4 people. The team will design, build, evaluate, and demonstrate an **end-to-end Big Data and AI solution** for a real-world financial problem. The team members will work collaboratively to complete the following assignments. At the first team meeting, the team should select the problem domain, use cases, and roles and responsibilities for each team member.

Select one of the following seven financial application domains (from A to G) and related use cases in section 2. **Implement one or more use cases, the more the better!**

Recommended project title:

“Big Data/AI-Powered **Domain Name System”** Replace **Domain Name** with the Name of the Problem Domain you choose from the following list.

2. Problem Domains

A. Financial Crime & Fraud Analytics

Use Cases:

- Credit Card Fraud Detection
- Fraud / Anomaly Detection
- Anti-Money Laundering (AML) Detection

Key Problems:

- Extremely imbalanced data (fraud < 0.1%)
- High false positives leading to customer friction
- Evolving fraud patterns and adversarial behavior

- Regulatory compliance requirements (FATF, Basel guidelines)

ML Value:

- Detection of complex behavioral anomalies (Bolton & Hand, 2002)
- Real-time transaction risk scoring (Bhattacharyya et al., 2011)
- Adaptive learning vs rule-based systems
- Graph/network analytics for AML (Weber et al., 2019)

Expected Business Impact:

- Reduced fraud losses
- Lower investigation costs
- Improved compliance outcomes
- Enhanced customer trust

B. Credit Risk & Default Intelligence

Use Cases:

- Loan Default Prediction / Credit Scoring
- Early Warning Signals for Loans or Investments

Key Problems:

- Incomplete borrower information
- Lagging indicators in traditional models
- Macroeconomic sensitivity
- Regulatory requirement for explainability (Basel II/III)

ML Value:

- Nonlinear risk modeling (Lessmann et al., 2015)
- Integration of alternative data sources (Bazarbash, 2019)
- Early deterioration detection
- Improved borrower segmentation

Expected Business Impact:

- Reduced default rates
- Improved credit approval precision
- Higher risk-adjusted returns
- Optimized capital allocation

C. Investment & Market Intelligence

Use Cases:

- Stock Return / Volatility Prediction
- Crypto Price Movement Prediction
- Financial News Sentiment Analysis
- Portfolio Rebalancing and Risk Monitoring

Key Problems:

- High noise and market randomness
- Non-stationary time series
- Information overload
- Difficulty linking signals to decisions

ML Value:

- Signal extraction from noisy data (Fama, 1970; Gu et al., 2020)
- NLP-based sentiment analysis (Tetlock, 2007)
- Dynamic risk modeling
- Pattern recognition across assets

Expected Business Impact:

- Improved alpha generation
- Enhanced risk-adjusted returns
- Faster response to market events
- Better portfolio allocation decisions

D. Customer Intelligence & Retention

Use Cases:

- Customer Churn Prediction

Key Problems:

- Silent churn without explicit signals
- High acquisition cost vs retention
- Fragmented customer data
- Limited behavioral visibility

ML Value:

- Predictive churn modeling (Verbeke et al., 2012)
- Customer segmentation and targeting
- Behavioral pattern detection
- Personalized retention strategies

Expected Business Impact:

- Increased customer lifetime value
- Reduced churn rate
- Improved cross-selling
- Higher engagement

E. Insurance Risk & Claims Analytics

Use Cases:

- Insurance Claim Risk Prediction

Key Problems:

- Fraudulent claims
- Information asymmetry
- Complex risk profiles
- Pricing inefficiencies

ML Value:

- Fraud detection in claims (Viaene et al., 2002)
- Risk-based pricing models
- Pattern recognition in claims data
- Automated underwriting

Expected Business Impact:

- Reduced claim losses
- Improved underwriting accuracy
- Operational efficiency
- Increased profitability

F. ESG, Compliance & Reputation Risk

Use Cases:

- ESG / Greenwashing Risk Scoring

Key Problems:

- Lack of standardized ESG data
- Greenwashing risk
- Regulatory pressure
- Inconsistent disclosures

ML Value:

- NLP analysis of ESG reports (Berg et al., 2022)
- Cross-source validation
- Anomaly detection in disclosures
- Automated ESG scoring

Expected Business Impact:

- Reduced reputational risk
- Improved ESG credibility
- Better sustainable investment decisions

G. Intelligent Advisory & Decision Support

Use Cases:

- Robo-Advisory Portfolio Recommendation
- Explainable AI for Analysts & Managers

Key Problems:

- Limited scalability of human advisory
- Black-box model concerns
- Complex decision environments
- Regulatory explainability requirements

ML Value:

- Automated portfolio optimization (Markowitz, 1952)
- Personalized recommendations
- Explainable AI methods (Ribeiro et al., 2016; Lundberg & Lee, 2017)
- Human-AI collaborative decision making

Expected Business Impact:

- Scalable advisory services
- Improved client trust
- Faster and better decisions
- Reduced reliance on manual analysis

Recommended Datasets:

• Domain	• Dataset	• Rows	• Target
• A. Fraud	• IEEE-CIS Fraud Detection	• 590K	• isFraud (3.5%)
• B. Credit	• Give Me Some Credit (GMSO)	• 150K	• SeriousDlqin2yrs (6.7%)
• B+	• UCI CC Default Clients	• 30K	• default.payment
• C Markets	• Optiver Realized Volatility	• 5M+	• realized_volatility
• C NLP	• Twitter Financial News Sentiment	• 11.9K	• Bearish/Bullish/Neutral
• D Churn	• KKBox WSDM 2018 Churn	• 2.3M	• is_churn (7% rate)
• E Insurance	• Porto Seguro Safe Driver	• 595K	• target (claim 3.6%)
• F ESG	• SEC EDGAR 10-K Risk Factors	• 1000s	• ESG/Climate NLP label
• G Advisory	• AmEx Default Prediction	• 458K	• target (default 25.9%)

3. Learning Objectives

By completing this project, students should be able to:

1. Apply the Big Data Machine Learning Framework taught in the class.
2. Define a financial business problem and translate it into a machine learning problem.
3. Perform data preparation, cleaning, exploration, and feature engineering.
4. Select suitable algorithms and justify model choices.
5. Apply cross-validation and hyperparameter tuning.
6. Compare baseline and improved models using appropriate metrics.

7. **Build a simple dashboard front end to demonstrate results.**
 8. **Use AI tools such as ChatGPT, Claude, Gemini, or GitHub Copilot responsibly.**
 9. **Maintain an AI-use audit trail throughout.**
 10. **Explain business implications, risks, limitations, and ethical concerns.**
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4. Required Technical Components

Minimum Model Requirements

Each team must implement at least:

1. **Baseline model**

- Linear regression, polynomial regression, decision tree, or naïve benchmark

2. **At least three advanced models**

- Random Forest
- Gradient Boosting
- XGBoost
- LightGBM
- SVM
- Neural Network
- KNN
- Logistic Regression (Classifier)
- Ridge/Lasso/Elastic Net
- Clustering
- Association

3. **Feature engineering**

- At least five meaningful engineered features or transformations
- Domain-specific justifications

4. **Hyperparameter tuning**

- At least one systematic approach: Grid Search, Random Search, or Bayesian Optimization

5. Cross-validation

- K-fold, stratified K-fold, or time-series split

6. Model interpretation

- Feature importance, SHAP, permutation importance, or partial dependence

7. Dashboard

- Full function demo – not static. Minimal: show the Machine Learning Framework and the visualization
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5. Dashboard Requirements (extra credits for extra features)

Students should use AI to build a usable system to demonstrate the value of their models:

Minimum Features

1. Data summary visualization
2. Model performance metrics
3. Prediction interface
4. Feature importance display
5. Business recommendation output

Example Tools

- Streamlit
 - Dash
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6. Final Presentation Format

Each team has **15 minutes for presentation + 5 minutes Q&A.**

Suggested structure:

1. Problem and business value — 2 minutes

2. Data and feature engineering — 3 minutes
3. Model development, cross-validation, & tuning — 4 minutes
4. Results and interpretation — 3 minutes
5. (Dashboard) Demo — 3 minutes

Suggested Presentation Focuses

1. Storytelling

- Clear narrative:
 - Problem → Approach → Solutions → Results → Impact

2. Focus on SMART Impacts

Not:

“Model accuracy is 95%.”

But:

“By optimizing recall, we reduce fraud loss by ~22% while increasing false alerts by only 5%, which is acceptable for compliance teams.”

3. Business Translation

You must answer:

- **What? Why?**
- **So what?**
- **Who** benefits?
- **What** action should be taken?

4. Data Empowered Decisions- Showcase what you learned this semester

- **How and Why:** Present the data processing, feature engineering, algorithms you have chosen, cross-validation, hyperparameter tuning, and validation.

5. Action-Oriented Recommendations - Examples:

- Suggest process modifications
- Set fraud detection thresholds
- Recommend credit approval cutoffs
- Suggest portfolio allocation adjustments

5. Dashboard Demonstration Quality

- Smooth live demo flow
- Clear explanation of:
 - Inputs
 - Outputs
 - Use cases
 - Performance

- Extra credits for additional features

Each student will grade and provide recommendations to all other teams. This will account for some of the participation grades. Insightful suggestions earn extra credit.

7. AI Tool Usage Policy

Students are allowed and encouraged to use AI tools (ChatGPT, Claude, GitHub Copilot, ...), but they must demonstrate responsible use of them.

Required AI Audit Trail

Each team must include a table **like** this and submit the project tracking documents:

Date	Task	AI Tool Used	Prompt Summary	AI Output Used?	Human Modification	Learning Reflection
Week 7	Feature engineering	ChatGPT	Asked for fraud detection feature ideas	Partially (1)	Removed unrealistic features	Learned that domain-specific features matter
Week 8	Streamlit dashboard	Claude Code	Asked for the dashboard structure	Yes (2)	Modified layout and charts	Learned app structure
Week 8	Hyperparameter tuning	ChatGPT	Asked for GridSearchCV example	Yes (3)	Adapted to LightGBM	Understood tuning parameters
	...					

Explain the AI tool used in more detail:

(1) ChatGPT suggested a, b, c, and we added velocity feature d, e, f, and removed unrealistic feature g, h, and l,...

(2) Claude suggested a layout and the following charts. We modified the layout to add A, B, and C and remove D because of ..., and added the following charts, ...

(3) ChatGPT suggested the following hyperparameters: we increased the learning rate to get the results faster, ...

Important Rule

AI-generated code must be reviewed, tested, and explained by students.

Students cannot simply submit AI-generated work without understanding it or modifying it.

8. Project Deliverables

A. Final Report

Suggested length: **20–30 pages**

Required sections:

1. Executive Summary

- Problem
- Solution
- Key findings
- Business value

2. Problem Definition

- Financial problem background
- Target users and pain points
- Business objectives
- Machine learning objectives

3. Data Description

- Dataset(s) source
- Data dictionary
- Target variable
- Key features
- Data quality issues

4. Exploratory Data Analysis

- Missing values
- Class imbalance
- Outliers
- Correlation analysis

- Important visual insights

5. Feature Engineering

- Original features
- Scaling or normalization
- Feature extraction and encoding
- Feature transformation
- New features created
- Feature selection
- Rationale for feature design

6. Model Development

- Baseline model
- Candidate algorithms
- Training strategy
- Cross-validation design
- Hyperparameter tuning method

7. Model Comparison

- Performance metrics
- Confusion matrix, ROC-AUC, PR-AUC, F1, RMSE, MAE, or other suitable metrics
- Baseline vs improved models
- Interpretation of results

8. Dashboard Demonstration

- Dashboard screenshots
- Key functions
- User workflow
- Business decision support value

9. AI Tool Usage Audit Trail

- Tools used
- Prompts submitted
- AI-generated suggestions
- Human modifications
- Errors found and corrected

- Evidence for learning and improvement

10. Ethics, Governance, and Risk

- Bias risk
- Privacy risk
- Explainability
- Model misuse risk
- AI tool dependency risk

11. Conclusion and Future Improvements

B. Dashboard Front End

Students should build a simple dashboard using AI-assisted coding.

Minimum dashboard requirements:

1. Upload or load test data
2. Display key dataset statistics
3. Show model predictions
4. Visualize model output and performance
5. Show important features
6. Provide a simple business interpretation

Suggested dashboard pages:

Page	Purpose
Home	Problem and solution overview
Data Explorer	Dataset summary and visualizations
Model Performance	Metrics, confusion matrix, ROC/PR curve
Prediction Demo	Input data and predicted output
Explainability	Feature importance or SHAP explanation
Business Recommendation	Suggested actions

C. Program Code Submission

In addition to the team final report and the final presentation, each team should submit the program code and AI usage tracking files.

All files should be placed in a subdirectory, and the directory should be compressed into a zip file named “DFS504_TeamX_Code.zip”, where X is your Team ID. Send the zip file to kyw@mail.nsysu.edu.tw.

- Project plan and team task assignments
- Dataset link or sample dataset
- Jupyter Notebook or Python scripts:
 - Data preprocessing code
 - Feature engineering code
 - Model training code
 - Hyperparameter tuning code
 - Model evaluation code
- Dashboard source code in a subdirectory of its own, with a Readme file with instructions for installation and usage

Recommended tool for coding:

- Python
- Pandas
- NumPy
- Scikit-learn
- XGBoost / LightGBM
- SHAP
- Plotly / Matplotlib
- Streamlit or Dash
- ChatGPT / Claude / Copilot

D. Final Submission Package

1. Final report PDF
2. Presentation slide file
3. AI audit trail document(s), including the AI prompts usage explanations, key decisions, versioning, modifications, and tool usage
4. The report and the presentation title pages should include the domain name, team members' student IDs, and their names.
5. Responsible AI Usage Declaration Form, including how AI tools are used and human contributions in each of the above components where AI is used.

All files should be placed in a subdirectory, and the directory should be compressed into a zip file named “DFS504_TeamX_Reports.zip”, where X is your Team ID. Send the code and report zip files to kyw@mail.nsysu.edu.tw.

9. Grading Rubric

Total: **100 points**

Category	Weight	
Problem Framing & Financial Relevance	15	Clarity of business problems, financial context, and ML translation
Data & Feature Engineering	15	Depth, creativity, and financial logic behind features
Model Development	10	Algorithm selection, baseline vs advanced models
Cross-validation, Hyperparameter Tuning & Performance Improvement	15	Rigor of tuning + measurable improvement
Dashboard & Usability	10 + 5	Functionality, clarity, and business usability
AI Audit Trail & Responsible Use	10	Transparency, learning evidence, and AI-human collaboration
Presentation & Business Insights	25	Clarity, storytelling, interpretation, and decision impact

Key Success Criteria (What Distinguishes A-Level Work)

Top teams will demonstrate:

1. Clear Performance Improvement

- Not just models → **show measurable gains**
- Example: F1 ↑, ROC-AUC ↑, RMSE ↓

2. Strong Feature Engineering

- **Domain-driven insights**

3. Intelligent Use of AI Tools

- Audit trails
- Evidence of iteration and refinement

4. Expected Business Impact

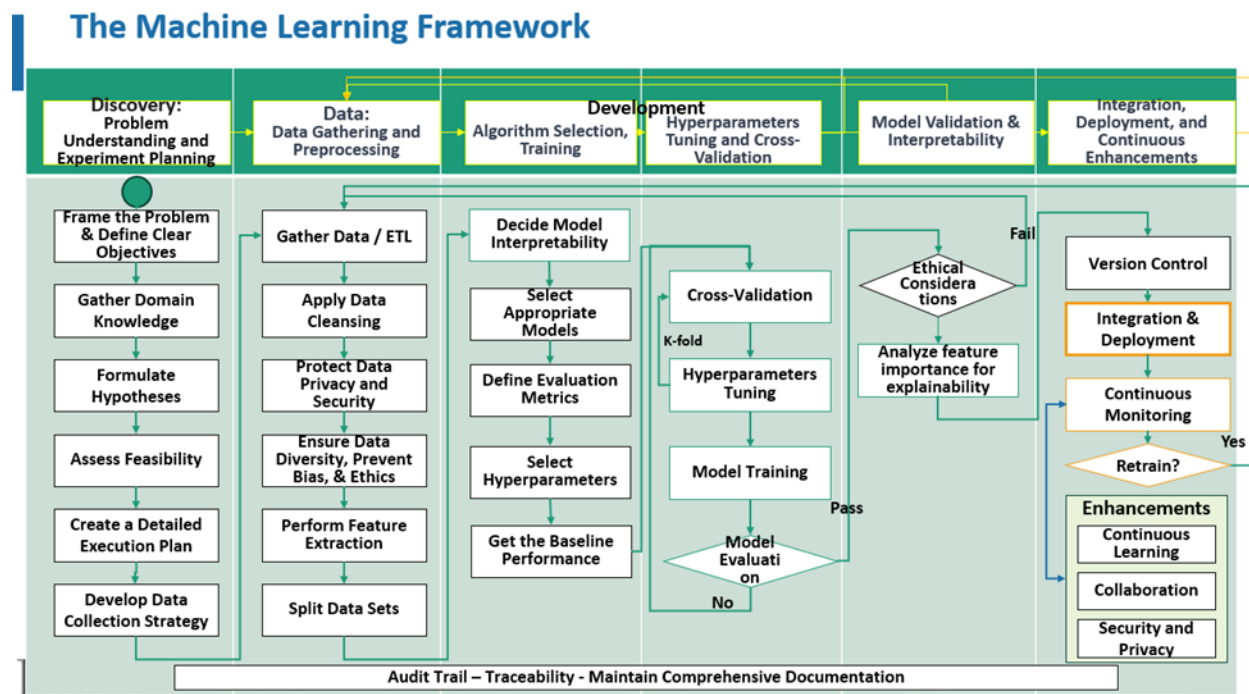
- Translate model output into:
 - Risk reduction
 - Profit improvement
 - Decision support

5. Working System

- Dashboard feels like a **real product**

10. Suggested Six-Week AI & Big Data Finance Project Schedule

“End-to-End Financial AI Solution with Dashboard & AI Audit Trail”



1. Framework Structure (Anchor for Entire Project)

Students must explicitly organize their work using this **Machine Learning Framework**:

Phase 1 — Discovery

- Problem definition (business + ML objective)
- Success metrics (business + technical SMART objectives)

Phase 2 — Data & Feature Engineering

- Data acquisition
- Cleaning, EDA
- Feature engineering

Phase 3 — Modeling

- Baseline model
- Advanced models
- Cross-validation

Phase 4 — Cross Validation and Hyperparameter Tuning

- Hyperparameter tuning
- Model comparison
- Performance improvement

Phase 5 — Deployment (Dashboard)

- Build front-end interface
- Integrate model outputs

Phase 6 — Governance & AI Audit

- AI tool usage tracking
- Bias, risk, explainability

2. Suggested Six-Week Timeline

Week 1 — Discovery + Data Setup - Define the problem and get data ready

Activities

- Select financial use case

- Define:
 - Business objective
 - ML objective
- Identify evaluation metrics
- Acquire dataset (Kaggle / APIs / open data)
- Initial EDA

Deliverables

- Problem statement (1 page)
 - Dataset description
 - Initial EDA notebook
 - AI audit trail starts
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Week 2 — Data Processing & Feature Engineering - Turn raw data into usable intelligence

Activities

- Data cleaning:
 - Missing values
 - Outliers
- Feature engineering:
 - Encoding
 - Scaling
 - Domain-driven features
- Feature selection (correlation, importance)

Deliverables

- Feature engineering report
 - Before vs after dataset comparison
 - Updated notebook
-

Week 3 and Week 4 — Baseline + Initial Models - Build working models quickly, then Hyperparameter Tuning & Optimization - Improve model performance rigorously

Baseline + Initial Models Activities:

- Build baseline model
- Train at least 3 models:
 - Logistic / Linear
 - Tree-based (RF, XGBoost, LightGBM)
 - One additional (SVM / NN / KNN)
- Apply cross-validation

Deliverables

- Baseline vs model comparison table
- Initial performance metrics
- Model selection rationale

Hyperparameter Tuning & Optimization Activities (Improve model performance rigorously)

- Hyperparameter tuning:
 - Grid Search / Random Search
- Compare tuned vs untuned models
- Handle:
 - Class imbalance (SMOTE, weighting)
 - Threshold optimization

Deliverables

- Tuning strategy explanation
- Performance improvement analysis
- Final model selection

Week 5 — Dashboard Development (AI-Assisted) - Turn model into usable business tool

Activities

- Use AI tools (ChatGPT / Claude Code) to:
 - Generate Streamlit/Dash app
- Build:
 - Prediction interface
 - Model performance visualization
 - Feature importance display

Deliverables

- Working dashboard
 - Screenshots + demo video
 - Code repository
-

Week 6 — Governance, AI Audit, Final Integration - Professional-level delivery

Activities

- Complete AI audit trail
- Add explainability (SHAP or feature importance)
- Risk & ethics analysis
- Final integration and testing

Deliverables

- Final report (20–25 pages)
- Dashboard demo
- AI audit log
- Final presentation (15 minutes)